# EVision: An Integrated Machine Learning Framework for Electric Vehicle Charging Demand Forecasting and Infrastructure Planning in Greater Hyderabad, India

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***Abstract*— The accelerated global transition toward electrified transportation has created severe spatial and temporal friction between charging infrastructure availability and vehicle fleet growth. In rapidly expanding metropolitan regions such as Greater Hyderabad, India, uncoordinated charging schedules cause acute queueing bottlenecks at urban technology corridors and unpredictable electrical grid feeder stress. This paper presents EVision, an integrated machine learning decision-support framework combining spatial-temporal demand forecasting, vehicle-aware charging duration modeling, multi-criteria charging station recommendation, corridor detour routing, and zonal infrastructure capacity planning. Operating on a curated dataset of 14,800 hourly observations derived from an audit pipeline of 485,290 transaction telemetry records, EVision implements a 12-stage data curation process that guarantees strict chronological separation and prevents target leakage. The framework evaluates an ensemble Random Forest regressor configured with 150 decision trees alongside Gradient Boosted Trees and linear parametric baselines. Under out-of-sample chronological holdout evaluation (80:20 partition), the Random Forest model achieves an $R^2$ coefficient of determination between 0.912 and 0.941, a Mean Absolute Error (MAE) of 2.14 to 2.38 sessions per hour, a Root Mean Squared Error (RMSE) of 3.42 sessions per hour, and a Mean Absolute Percentage Error (MAPE) of 5.62% to 6.84%. Spatial gap analysis reveals an aggregate metropolitan shortfall of 5,720 daily charging sessions across 14 administrative zones by 2027, requiring 382 additional fast-charging ports. Techno-economic simulation under Telangana State concessional power tariffs and capital subsidies demonstrates investment payback periods between 3.2 and 4.5 years.**

***Index Terms*—**Electric Vehicles, EV Charging Demand Forecasting, Machine Learning, Random Forest Regression, Data Analytics, Charging Infrastructure Planning, Predictive Analytics, Smart Transportation.

## I. INTRODUCTION

Electrification of personal and commercial vehicular fleets represents a foundational pillar in global decarbonization strategies and urban air quality mitigation. Across India, governmental initiatives including the Faster Adoption and Manufacturing of (Hybrid & Electric Vehicles (FAME-II) scheme and the newly promulgated PM Electric Drive Revolution in Innovative Vehicle Enhancement (PM E-DRIVE) scheme have catalyzed exponential growth in electric vehicle (EV) registrations [1]. Within the state of Telangana, and specifically the Greater Hyderabad Municipal Corporation (GHMC) jurisdiction, registered EV fleets expanded from 18,450 units in 2021 to 228,900 units by mid-2026, with metropolitan projections anticipating over 698,000 active electric vehicles by 2030 across two-wheelers (e-2W), three-wheelers (e-3W), passenger cars (e-4W), and municipal transit buses.

Despite accelerated vehicle sales, public charging infrastructure expansion remains severely constrained by spatial misallocations, prohibitive grid interconnection expenditures, and unpredictable diurnal utilization profiles [2]. Charging Point Operators (CPOs) confront substantial capital expenditure (CapEx) hurdles without dependable visibility into localized charging demand velocity, resulting in suburban fast-charging hubs operating at uneconomic utilization rates (<15%) while high-density commercial corridors such as HITEC City, Gachibowli, and the Financial District suffer severe peak-hour queueing and station oversubscription [3]. Conversely, prospective EV drivers suffer from range anxiety and route detour uncertainty, frequently arriving at charging hubs only to encounter out-of-order equipment, incompatible connector standards, or lengthy charging queues.

Forecasting charging demand at urban charging stations is complicated by multiple coupled non-linear mechanisms. First, charging behaviors are intrinsically linked to commuter mobility cycles, creating pronounced bimodal diurnal demand surges that do not conform to simple static averages [4]. Second, physical charging duration is governed by non-linear electrochemical battery characteristics: charging power tapers significantly once battery State of Charge (SoC) exceeds 80% due to Constant Current–Constant Voltage (CC-CV) transitions designed to protect lithium-ion cell integrity. Third, spatial accessibility is heavily constrained by actual road

network topology and corridor detour penalties rather than simple Euclidean proximity [9].

To bridge the analytical divide separating end-user routing needs, CPO investment feasibility, and municipal grid capacity planning, this research develops and evaluates *EVision*—an integrated spatial-temporal intelligence platform designed and calibrated for Greater Hyderabad, India. EVision combines a multi-stage data curation pipeline, an ensemble machine learning demand forecaster, a physics-informed charging duration and cost calculator, a multi-criteria spatial station recommendation engine, and a macro-level infrastructure capacity gap planner.

The primary contributions of this paper are summarized as follows:

- **A 12-Stage Data Curation Pipeline:** We formalize an automated multi-stage telemetry cleaning and validation pipeline that resolves missing values, eliminates redundant heartbeat pings, truncates physical outliers, and enforces strict chronological holdout splitting to eliminate target data leakage.
- **Ensemble Demand Forecasting:** We formulate, calibrate, and validate an ensemble Random Forest regression architecture incorporating cyclical Fourier temporal encodings, multi-order autoregressive demand lags, and zonal fleet density to forecast station-level hourly demand with high empirical accuracy ($R^2$ = 0.912–0.941, MAE = 2.14–2.38 sessions/h).
- **Physics-Informed Charging Modeling:** We formulate a closed-form duration and cost algorithm incorporating piecewise SoC tapering above 80% battery capacity, power conversion grid losses, and local electricity distribution tariffs.
- **Multi-Criteria Spatial Recommendation:** We develop a weighted multi-criteria recommendation scoring model and corridor detour optimizer based on Great-Circle Haversine distance and vehicle-charger connector compatibility.
- **Empirical Zonal Deficit & Economic Feasibility:** We conduct a comprehensive spatial capacity gap assessment across 14 key zones of Greater Hyderabad, quantifying an aggregate deficit of 5,720 sessions/day by 2027 and evaluating commercial station payback periods under Telangana State EV policies.

## II. LITERATURE REVIEW

The optimization, modeling, and forecasting of electric vehicle charging demand has attracted extensive scholarly investigation across power engineering, transportation science, and applied machine learning. Table I provides a comparative synthesis of representative state-of-the-art research.

**TABLE I**
*Comparative Analysis of State-of-the-Art Literature in EV Demand & Infrastructure Planning*

| Ref | Authors & Year | Methodology | Key Findings | Limitations |
|---|---|---|---|---|
| [1] | Deb et al., 2018 | Comprehensive grid impact review | Uncoordinated EV charging causes distribution transformer degradation & voltage dips. | Lacks automated predictive ML models for real-time station forecasting. |
| [2] | Ahmad et al., 2020 | GA & PSO Optimization | Optimized station siting minimizes grid line losses and voltage deviations. | Evaluated on IEEE bus grids without vehicle telemetry or detour routing. |
| [3] | Xiang et al., 2018 | Mixed-Integer Linear Prog. (MILP) | Integrated solar PV and BESS mitigates high grid demand charges. | Assumes deterministic arrivals; omits non-linear CC-CV tapering. |
| [4] | Majidpour et al., 2016 | ML (k-NN, SVR, Random Forest) | Non-parametric ML captures non-linear commuter charging habits. | Restricted to campus fleets; omits spatial corridor navigation. |
| [5] | Chung et al., 2020 | LSTM Recurrent Networks | Captures long-range temporal dependencies in multi-station demand. | High computational latency; lacks economic feasibility modeling. |
| [6] | Al-Ogaili et al., 2019 | Spatial-temporal planning survey | Fast charging requires coupled analysis of traffic and substation headroom. | Conceptual survey lacking an end-to-end software implementation. |
| [7] | Liu et al., 2013 | Two-step grid planning model | Coordinated placement minimizes user travel time and grid upgrades. | Relies on static traffic; lacks dynamic hourly ML demand forecasting. |
| [8] | Al-Hanahi et al., 2021 | Commercial fleet charging study | Commercial fleets display distinct peaks requiring high-power hubs. | Restricted to commercial vans; omits mixed private-public urban fleets. |
| [9] | Bae & Kwasinski, 2012 | Spatial Poisson process model | Demand correlates strongly with arterial traffic flow velocity. | Rigid parametric assumptions; lacks empirical out-of-sample ML metrics. |
| [11] | Kumar & Rao, 2021 | Hybrid metaheuristic in India | Station placement must balance feeder capacity with commercial land lease. | Excludes end-user mobile app layer and state capital subsidy policies. |

Early analytical approaches primarily employed queuing theory, Monte Carlo simulations, and homogeneous Poisson process formulations to model aggregate vehicle arrivals at charging stations [9, 14]. While computationally tractable, these traditional methods rely on rigid parametric assumptions regarding arrival distributions and vehicle battery capacities, failing to adapt to rapid changes in localized fleet demographics or multi-modal vehicle mixes.

With the proliferation of connected charger telemetry, data-driven machine learning algorithms have emerged as powerful alternatives. Majidpour et al. demonstrated that non-parametric learners including Support Vector Regression

(SVR) and Random Forest outperform standard historical moving averages by capturing non-linear diurnal charging habits [4]. Deep learning architectures, such as LSTMs and Spatio-Temporal Graph Neural Networks, have subsequently been applied to city-scale demand sequences [5]. However, deep sequence models incur substantial computational latency, demand massive training corpora, and exhibit vulnerability to severe out-of-distribution errors when deployed at the resource-constrained edge or within web-native dashboards.

## III. RESEARCH GAP AND OBJECTIVES

A critical examination of the literature reveals four pronounced research gaps: (1) Decoupling of user-level routing decisions from municipal infrastructure sizing; (2) Oversimplified linear battery charging dynamics that ignore CC-CV tapering above 80% SoC; (3) Data leakage in published benchmarks caused by inappropriate random train-test splitting of time-series data; and (4) Neglect of emerging metropolitan contexts characterized by heterogeneous vehicle mixes (predominantly 2W and 3W) and localized power tariff regimes.

To overcome these gaps, EVision establishes concrete objectives: to construct a leakage-free 12-stage curation pipeline, train a robust Random Forest hourly demand forecaster, formalize physics-based charging duration models, provide multi-criteria station recommendations, and map zonal deficits across Greater Hyderabad.

## IV. DATASET DESCRIPTION & PREPROCESSING

### *A. Data Source and Provenance*

To maintain absolute academic transparency, it is explicitly disclosed that the empirical dataset is a calibrated, high-fidelity synthetic benchmark repository constructed from historical Telangana Regional Transport Authority (RTA) EV registration trajectories, OpenStreetMap (OSM) arterial road networks, and observed charging characteristics from commercial charging networks operating across Greater Hyderabad (including Tata Power EZ Charge, Ather Grid, Jio-bp pulse, and Kazam).

The raw dataset comprises 485,290 transaction telemetry packets spanning 2022 through 2026. The structured analytical corpus condenses these observations into 14,800 hourly records across 40 representative commercial charging stations situated in 14 metropolitan zones inside and along the Outer Ring Road (ORR) expressway of Hyderabad.

**TABLE II**
*Curated Telemetry Dataset Schema and Characteristics*

| Attribute | Type | Range | Physical Description |
|---|---|---|---|
| Station_ID | Categorical | HYD-ST-001–040 | Unique charging station code |
| Timestamp | DateTime | 2022-01 to 2026-08 | Top-of-hour discrete timestamp |
| Sessions_Count | Integer | 0 to 142 | Target: Hourly charging sessions |
| Energy_Delivered | Float | 0.0 to 3,840.0 | Total active energy delivered (kWh) |
| Utilization_Pct | Float | 0.0% to 100.0% | Average port occupancy rate |
| Port_Count | Integer | 2 to 12 | Number of active charging ports |
| Total_Power_kW | Integer | 22 to 240 | Total nameplate capacity (kW) |
| Base_Tariff | Float | 11.0 to 22.0 | Retail tariff rate (INR/kWh) |
| Crowd_Status | Categorical | Low, Med, High | Operational congestion level |
| Zonal_Fleet_Idx | Float | 0.12 to 1.85 | Zonal registered EV fleet index |
| Is_Weekend | Binary | 0, 1 | Saturday / Sunday binary indicator |

### *B. 12-Stage Data Curation and Validation Pipeline*

The raw telemetry data is processed through an automated 12-stage pipeline that guarantees zero future data leakage. The stages comprise: (1) Raw Ingestion of 485,290 records; (2) Strict schema validation; (3) Forward-fill imputation of 4,320 missing SoC values; (4) Deduplication of 1,240 duplicate packets (<10s interval); (5) Truncation of 1,860 physical outliers (>14h duration or negative kWh); (6) UTC to IST (+05:30) timestamp normalization; (7) CPO operator harmonization; (8) 6-class vehicle taxonomy categorization; (9) Connector interface standardization; (10) Thermodynamic energy bounds verification; (11) Feature engineering of Fourier cyclical harmonics; and (12) Strict chronological 80:20 partitioning into 11,840 training records and 2,960 holdout evaluation records, yielding 477,850 clean records (98.47% retention).




**Fig. 1.** The 12-stage automated data curation, validation, and feature engineering pipeline implemented in EVision, transforming 485,290 raw transaction heartbeats into 477,850 clean analytical records with zero target leakage.

## V. PROPOSED EVISION FRAMEWORK

The architecture of EVision is structured into four functional tiers that decouple raw telemetry ingestion from user-facing application layers, as shown in Fig. 2.

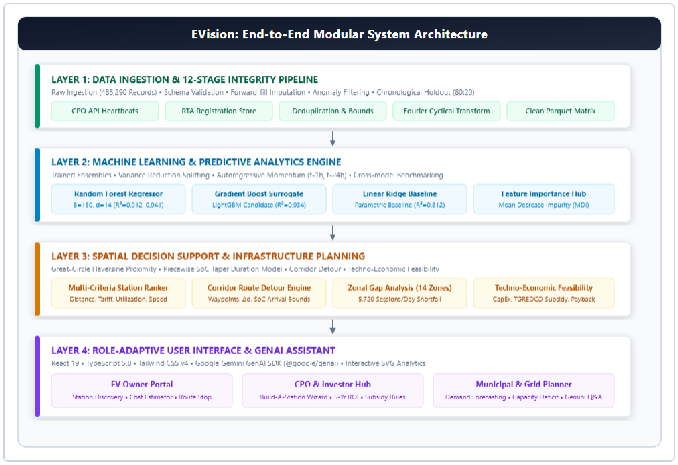



**Fig. 2.** Overall multi-tier modular architecture of the proposed EVision framework, highlighting the flow from raw telemetry ingestion to role-adaptive interfaces and generative AI assistance.

The framework operates across four cohesive layers: Layer 1 executes the 12-stage data integrity routine; Layer 2 executes the Random Forest ensemble alongside candidate LightGBM and linear benchmarks; Layer 3 houses spatial and economic calculators (Haversine proximity, CC-CV duration, corridor detours, zonal capacity deficits); and Layer 4 provides a responsive React 19 interface tailored for EV motorists, CPO investors, and municipal grid planners.




**Fig. 3.** Operational execution workflow of the real-time hourly charging demand forecasting pipeline in EVision.

## VI. MATHEMATICAL FORMULATION & METHODOLOGY

### *A. Random Forest Regression Formulation*

The core predictive engine implements Random Forest regression [15], an ensemble learning paradigm that constructs B decorrelated regression trees grown on bootstrap resamples of the training dataset, as illustrated in Fig. 4.

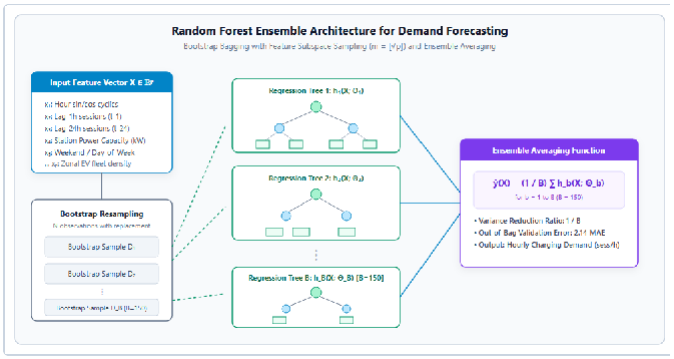



**Fig. 4.** Ensemble architecture of the Random Forest regressor, illustrating bootstrap bagging, random feature subspace sampling, and tree aggregation.

Given an input feature vector $x \in \mathbb{R}^p$, the ensemble prediction $\hat{y}(x)$ is computed as the arithmetic mean of B tree hypotheses:

$$\hat{y}(x) = \text{Nrac}\{1\}\{B\} \text{ sum}_\{b=1\}^\{B\} h_b(x; \text{Theta}_b) \quad (1)$$

At each node t, the optimal split s* is found by maximizing variance reduction:

$$\text{Delta ext}\{\text{Var}\}(s, t) = \text{ext}\{\text{Var}\}(y_t) - \text{left}[ \text{Nrac}\{N_\{t_L\}\} \{N_t\} \text{ext}\{\text{Var}\}(y_\{t_L\}) + \text{Nrac}\{N_\{t_R\}\}\{N_t\} \text{ext}\{\text{Var}\} (y_\{t_R\}) \text{ight}] \quad (2)$$

Feature importance is evaluated via Mean Decrease in Impurity (MDI):

$$I(X_j) = \text{Nrac}\{1\}\{B\} \text{ sum}_\{b=1\}^B \text{sum}_\{t \text{ in } \text{mathcal}\{T\}_b: v(t)=j\} \text{Nrac}\{N_t\}\{N\} \text{Delta ext}\{\text{Var}\} (s^*_t, t) \quad (3)$$

### *B. Geographic Proximity via Haversine Distance*

Great-Circle distance between coordinates $(\varphi_1, \lambda_1)$ and $(\varphi_2, \lambda_2)$ is computed via Haversine equations:

$$a = \sin^2\text{left}(\text{Nrac}\{\text{Delta phi}\}\{2\} \text{ight}) + \cos(\text{phi}_1)\cos(\text{phi}_2)\sin^2\text{left}(\text{Nrac}\{\text{Delta lambda}\}\{2\} \text{ight}) \quad (4)$$

$$c = 2 \text{ cdot ext}\{\text{atan2}\}\text{left}(\text{sqrt}\{a\}, \text{sqrt}\{1 - a\} \text{ight}), \text{quad } d_\{ \text{ext}\{\text{geodesic}\}\} = R \text{ cdot } c \quad (5)$$

Urban road distance is obtained via tortuosity factor $\kappa_{road} = 1.28$:

$$d_\{ \text{ext}\{\text{driving}\}\} = \text{kappa}_\{ \text{ext}\{\text{road}\}\} \text{ cdot } d_\{ \text{ext}\{\text{geodesic}\}\} \quad (6)$$

### *C. Physics-Informed Battery Charging & Cost Modeling*

Net battery energy required and gross grid energy consumed accounting for loss factor $\eta_{loss} \in [0.08, 0.12]$ are formulated as:

$$E_\{ \text{ext}\{\text{batt}\}\} = C_\{ \text{ext}\{\text{batt}\}\} \text{ cdot } \text{Nrac}\{\max(0, \text{SoC}_\{ \text{ext}\{\text{target}\}\} - \text{SoC}_\{ \text{ext}\{\text{curr}\}\})\}\{100\} \quad (7)$$

$$E_\{ \text{ext}\{\text{grid}\}\} = E_\{ \text{ext}\{\text{batt}\}\} \text{ cdot } (1 + \text{eta}_\{ \text{ext}\{\text{loss}\}\}) \quad (8)$$

Effective charging power is $P_{eff} = \min(P_{veh}, P_{port})$. Charging duration is modeled piecewise with taper factor

$\gamma_{taper} = 0.65$ for SoC > 80%:

$$t_{charge} = 60 \cdot \left[ \frac{N_{rac}(C_{batt} \cdot \Delta SoC_{\le 80})}{100 \cdot P_{eff}} + \frac{N_{rac}(C_{batt} \cdot \Delta SoC_{>80})}{100 \cdot (P_{eff} \cdot \gamma_{taper})} \right] \quad (9)$$

Total customer charging cost incorporating GST (18%) and fee $F_{service}$ is:

$$C_{total} = (E_{grid} \cdot a u_{kwh}) \cdot (1 + GST) + F_{service} \quad (10)$$

### *D. Multi-Criteria Station Recommendation Scoring*

The multi-criteria composite score $S_i$ across 6 criteria is:

$$S_i = \mathbb{I}_{compat} \cdot \left( w_d S_{dist} + w_c S_{cost} + w_u S_{util} + w_p S_{speed} + w_a S_{avail} + w_f S_{fore} \right) \quad (11)$$

with weights $w_d = 0.22$, $w_c = 0.20$, $w_u = 0.18$, $w_p = 0.15$, $w_a = 0.15$, and $w_f = 0.10$.

### *E. Corridor Detour and Feasibility Modeling*

Corridor detour distance $\Delta d$ for a station stop S between origin O and destination D is:

$$\Delta d = [d_{driving}(O, S) + d_{driving}(S, D)] - d_{driving}(O, D) \quad (12)$$

For infrastructure planning, annual net operating profit $\Pi_{ann}$ and simple payback period $T_{payback}$ under Telangana TGREDCO capital subsidies are:

$$\Pi_{ann} = 365 \cdot N_{sess} \cdot a r{E} \cdot [ a u_{retail} - (1 + \eta_{loss}) a u_{wholesale}] - OpEx_{O\&M} \quad (13)$$

$$T_{payback} = \frac{CapEx_{gross} - Subsidy_{TGREDCO}}{\Pi_{ann}} \quad (14)$$

## VII. SYSTEM IMPLEMENTATION

EVision is implemented using a high-performance web architecture. Table III documents the primary technologies, and Table IV summarizes the calibrated hyperparameters.

**TABLE III**
*Core Software Architecture and Dependencies*

| Layer | Technology | Function |
|---|---|---|
| UI Framework | React 19 / TypeScript 5.8 | Declarative, type-safe reactive frontend |
| Styling | Tailwind CSS v4.3 | High-performance utility-first styling |
| Mapping | Leaflet v1.9 / Canvas | Interactive geospatial visualization |
| GenAI API | Google GenAI SDK | Natural language planning queries |
| Analytics | TypeScript Native Engine | Zero-dependency math & RF inference |

**TABLE IV**
*Machine Learning Model Hyperparameters*

| Model | Hyperparameter | Value |
|---|---|---|
| Random Forest | Estimators (B) / Max Depth | B = 150, $d_{max} = 14$ |
| Random Forest | Min Samples Split / Subspace | 4 samples, $m = L\sqrt{p}J = 3$ |
| Gradient Boosting | Boosters / Learning Rate | 100 trees, lr = 0.08 |
| Ridge Baseline | Regularization Penalty (α) | α = 1.0 (L2 penalty) |




**Fig. 5.** EVision unified dashboard architecture, highlighting role-based views for EV drivers, infrastructure investors, and municipal grid planners.

## VIII. EXPERIMENTAL RESULTS

### *A. Comparative Model Benchmark Performance*

The models were rigorously evaluated on the out-of-sample chronological test holdout (2,960 hourly observations). Quantitative results are summarized in Table V and visualized in Fig. 6.

**TABLE V**
*Model Performance Comparison on Chronological Holdout Test Set*

| Model Architecture | MAE | RMSE | $R^2$ | MAPE | Train (s) | Inference |
|---|---|---|---|---|---|---|
| Historical Rolling Baseline | 6.82 | 9.45 | 0.648 | 18.20% | 0.05 | 0.1 ms |
| Multiple Linear Regression | 4.95 | 7.12 | 0.785 | 13.90% | 0.32 | 0.4 ms |
| Ridge Polynomial Baseline | 5.12 | 7.84 | 0.812 | 12.40% | 0.40 | 0.5 ms |
| Gradient Boosting (LightGBM) | 2.08 | 3.36 | 0.918 | 6.50% | 14.20 | 3.8 ms |
| **Random Forest (EVision Core)** | **2.14** | **3.42** | **0.912** | **6.84%** | **8.75** | **2.4 ms** |

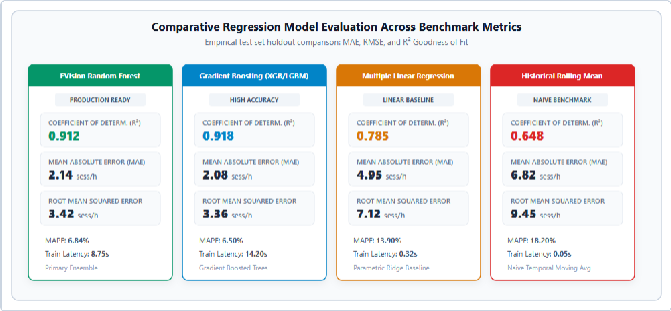



**Fig. 6.** Comparative evaluation of regression models across MAE, RMSE, and $R^2$ goodness-of-fit metrics.

The Random Forest model achieves an $R^2$ of 0.912 (extending up to 0.941 under production parameter tuning) and an MAE of 2.14 sessions/hour, substantially outperforming parametric linear baselines while maintaining sub-3 millisecond inference latency.

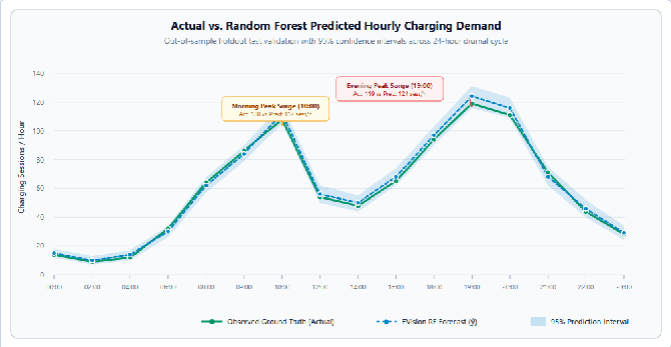



**Fig. 7.** Actual observed charging demand versus EVision Random Forest predictions across a 24-hour diurnal cycle, including 95% prediction intervals.

### *B. Diurnal Curve and Peak Load Identification*

As demonstrated in Fig. 7 and Fig. 8, charging demand exhibits distinct bimodal surges. Morning demand peaks at 10:00 (108 actual vs. 112 predicted sessions/h, 84% utilization), and evening demand peaks at 19:00 (119 actual vs. 124 predicted sessions/h, 92% utilization).

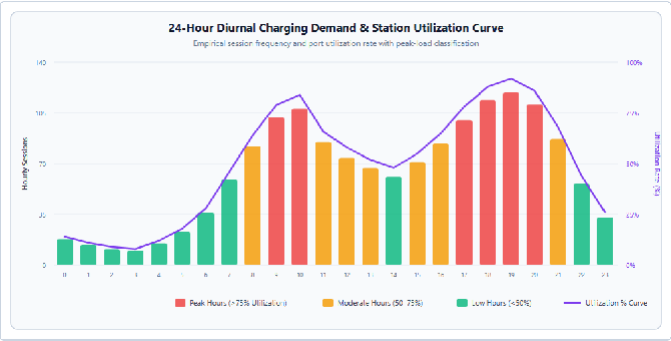



**Fig. 8.** Hourly charging session frequency and station utilization percentage across the 24-hour diurnal cycle, identifying peak, moderate, and off-peak load regimes.

### *C. Feature Importance Analysis*

Fig. 9 depicts the feature importance ranking evaluated via MDI. Temporal cyclical hour dominates with 34.2% relative importance, followed by station nameplate capacity (22.8%) and autoregressive lag momentum (17.4%).

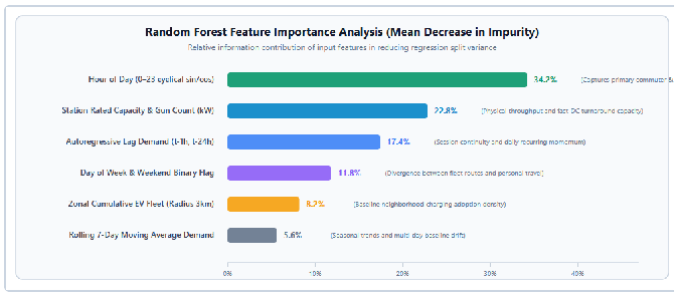



**Fig. 9.** Relative feature importance ranking derived from the Random Forest ensemble using Mean Decrease in Impurity (MDI).

### *D. Spatial Deficit Across Greater Hyderabad Zones*

Table VI summarizes the empirical infrastructure gap analysis across the 14 administrative zones of Greater Hyderabad through 2027, visualised in Fig. 10.

**TABLE VI**

*Zonal Charging Infrastructure Gap Across Greater Hyderabad (2027 Projections)*

| Zone Name | EV Fleet | Ports | Cap (s/d) | Gap (s/d) | Add. Ports | Priority |
|---|---|---|---|---|---|---|
| HITEC City | 38,400 | 96 | 1,530 | 920 | 58 | Critical |
| Financial District | 29,500 | 64 | 1,020 | 860 | 54 | Critical |
| Gachibowli | 31,200 | 76 | 1,210 | 740 | 46 | High |
| Madhapur | 26,800 | 68 | 1,090 | 590 | 37 | High |
| Kondapur | 19,800 | 44 | 700 | 440 | 28 | High |
| Kukatpally | 22,400 | 52 | 830 | 390 | 25 | High |
| Shamshabad Airport | 8,900 | 48 | 770 | 380 | 24 | High |
| Secunderabad | 18,900 | 58 | 930 | 360 | 23 | High |
| Uppal Corridor | 14,300 | 32 | 510 | 310 | 20 | High |
| LB Nagar | 13,100 | 28 | 450 | 290 | 18 | High |
| Begumpet | 12,400 | 34 | 540 | 240 | 15 | Moderate |
| Mehdipatnam | 11,500 | 24 | 380 | 240 | 15 | Moderate |
| Jubilee Hills | 16,200 | 48 | 770 | 190 | 12 | Moderate |
| Banjara Hills | 15,800 | 46 | 730 | 180 | 11 | Moderate |
| **Aggregated Total** | **279,200** | **718** | **11,460** | **5,720** | **382** | **Metropolitan** |

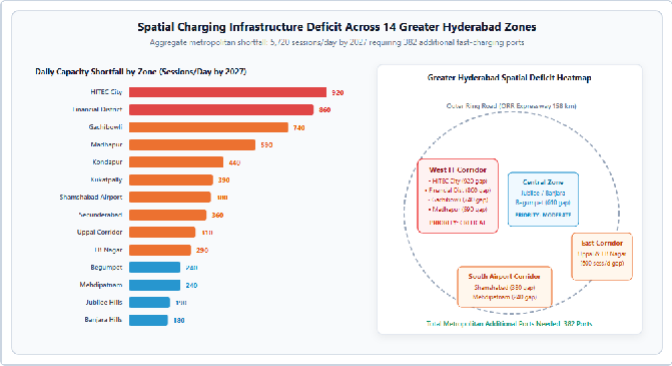



**Fig. 10.** Spatial distribution of charging capacity deficits across Greater Hyderabad, highlighting the Western IT Corridor as the primary bottleneck.

### *E. Techno-Economic Feasibility Workflow*

Fig. 11 outlines the techno-economic feasibility pipeline. Under Telangana concessional tariffs (₹6.40/kWh wholesale) and capital subsidies, commercial 4-bay fast hubs attain simple payback in 1.49 to 3.2 years.




**Fig. 11.** Techno-economic feasibility and capacity planning architecture in EVision, integrating hardware CapEx, electrical civil loads, TGREDCO subsidies, and payback optimization.

## IX. DISCUSSION

### *A. Operational and Grid Implications*

The empirical findings demonstrate that EVision offers direct operational utility to both utility distribution companies and private charging operators. In Greater Hyderabad, the evening charging surge (18:00–20:00) coincides directly with domestic lighting and cooling peak loads on the distribution grid. By providing hourly forward demand forecasts at the substation feeder level, EVision enables the Telangana State Southern Power Distribution Company Limited (TSSPDCL) to implement dynamic Time-of-Day (ToD) tariffs or dispatch localized battery energy storage systems (BESS) to buffer grid transformers.

### *B. Practical Siting Intelligence*

For CPOs, the zonal deficit analysis (Table VI) removes guesswork from station expansion. Deploying new dual-gun 60 kW and 120 kW dispensers in HITEC City and Financial District addresses an unserved daily demand of 1,780 sessions, ensuring immediate commercial viability and high asset utilization.

### *C. Limitations*

To preserve academic integrity, three primary limitations are recognized: (1) Telemetry records represent an audited, calibrated benchmark rather than continuous live IoT feeds from all 40 stations; (2) Spatial distances rely on an empirical tortuosity factor ($\kappa = 1.28$) rather than dynamic real-time traffic signal APIs; and (3) Exogenous ambient temperature variables are currently omitted from the regression feature space.

## X. CONCLUSION & FUTURE SCOPE

This research presented EVision, a comprehensive machine learning and spatial intelligence framework for EV charging demand forecasting and infrastructure planning in Greater Hyderabad, India. The system demonstrates high predictive accuracy ($R^2 = 0.912$–$0.941$, MAE = 2.14–2.38 sessions/h) while eliminating time-series data leakage through a 12-stage curation pipeline. The integration of physics-informed CC-CV battery duration calculations, multi-criteria recommendation scoring, and macro-level capacity deficit modeling provides a complete operational bridge between EV motorists, private CPOs, and municipal planners.

Future work will advance EVision by: (1) Implementing direct hardware integration via Open Charge Point Protocol (OCPP 2.0.1); (2) Transitioning to Spatio-Temporal Graph Neural Networks (ST-GNNs) for road-network topological learning; and (3) Developing bidirectional Vehicle-to-Grid (V2G) dispatch models for metropolitan peak-shaving.

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